OsdagBridge Report — Visual Enhancements

Nov 13, 2026 · Developers

Beyond the six required tasks, the report now has one consistent look, reads correctly across page breaks, and never shows a misleading value. Every "Before" image is from the original code (commit `d74f56d`) and every "After" image from the final code, on the same sample bridge.

At a glance

Enhancement	What the reader now sees	Where
Continued tables	"Table X.Y (continued)" with the column headings on every new page	All multi-page tables
Groups kept whole	Each girder's or bracing pair's rows stay on one page with its label	Chapters 2 and 5
Brand-coloured headings	Chapter, section and subsection headings in dark Osdag green (5.6:1 contrast)	Whole report
Table header band	Light green band behind every table header row, including merged headers	Whole report
Consistent captions	Bold "Table X.Y:" captions set in one place; capitalised "Parameter / Value" headers	Whole report
PASS / FAIL colouring	Failing values in the same red in tables and graphs, always with a text label	Chapter 5, graphs
Graphs in the report's font	Charts use Computer Modern, matching the body text, with units on every axis	Figures 5.1–5.2, 7.1–7.2
Missing shown as missing	"Not calculated" or "No force from analysis" instead of a misleading 0	Graphs, Section 5.4
Correct page furniture	Black footer text, correct "Page X of Y", symbols like ≤ printed	Every page

Tables that read across pages

A reader landing on any page of a long table now knows which table it is and what each column means.

- **Before:** Table 5.1 continued onto page 21 with no headings. Girder G4's rows were split from its label, and the table ran down to the footer.
- **After:** the page opens with "Table 5.1 (continued)" and the shaded heading row. Each girder's rows stay together, and the page ends with "Continued on next page".

BEFORE

	Gross Area, A (cm²)	387.0
	Moment of Inertia, I_z (cm⁴)	1881957.85
	Elastic Section Modulus, Z_{ez} (cm³)	22538.42
	Plastic Section Modulus, Z_{pz} (cm³)	25100.25
	Effective Slab Width, b_{eff} (mm)	2430.0
	Transformed Composite I_z (cm⁴)	4337628.66
	Depth to Plastic Neutral Axis (mm)	261.2
G3	Depth, D (mm)	1670.0
	Top Flange Width, b_f (mm)	510.0
	Bottom Flange Width, b_f (mm)	510.0
	Top Flange Thickness, t_f (mm)	22.0
	Bottom Flange Thickness, t_f (mm)	22.0
	Web Thickness, t_w (mm)	10.0
	Gross Area, A (cm²)	387.0
	Moment of Inertia, I_z (cm⁴)	1881957.85
	Elastic Section Modulus, Z_{ez} (cm³)	22538.42
	Plastic Section Modulus, Z_{pz} (cm³)	25100.25
	Effective Slab Width, b_{eff} (mm)	2430.0
	Transformed Composite I_z (cm⁴)	4337628.66
	Depth to Plastic Neutral Axis (mm)	261.2
G4	Depth, D (mm)	1670.0
	Top Flange Width, b_f (mm)	510.0
	Bottom Flange Width, b_f (mm)	510.0
	Top Flange Thickness, t_f (mm)	22.0
	Bottom Flange Thickness, t_f (mm)	22.0
	Web Thickness, t_w (mm)	10.0
	Gross Area, A (cm²)	387.0

AFTER

Table 5.1 (continued)

Girder	Property	Value
G2	Depth, D (mm)	1670.0
	Top Flange Width, b_f (mm)	510.0
	Bottom Flange Width, b_f (mm)	510.0
	Top Flange Thickness, t_f (mm)	22.0
	Bottom Flange Thickness, t_f (mm)	22.0
	Web Thickness, t_w (mm)	10.0
	Gross Area, A (cm²)	387.0
	Moment of Inertia, I_z (cm⁴)	1881957.85
	Elastic Section Modulus, Z_{ez} (cm³)	22538.42
	Plastic Section Modulus, Z_{pz} (cm³)	25100.25
	Effective Slab Width, b_{eff} (mm)	2430.0
	Transformed Composite I_z (cm⁴)	4337628.66
	Depth to Plastic Neutral Axis (mm)	261.2

Continued on next page

Figure 1 — Table 5.1 on page 21, before and after.

One consistent look

Every heading, table header and caption now follows the same rules, set once in `styles.py`.

- **Headings:** chapter, section and subsection headings use a dark Osdag green. The brand's lighter green (#91B014) is only 2.5:1 on white, too faint for text, so headings use #5A700D (5.6:1) and the lighter green stays on the header and footer rules.
- **Table headers:** a light green band (#EEF3DC) sits behind every header row, so headings stand apart from data at a glance. Black text on it keeps 18:1 contrast.
- **Captions and labels:** table captions are bold through one setting rather than hand-written bold in each caption. Lowercase "parameter / value" headers are now capitalised like the rest.
- **Load tables:** Chapter 3's vehicle and footway loads are now separate tables. Rows that showed "N/A" or a blank now show calculated values.

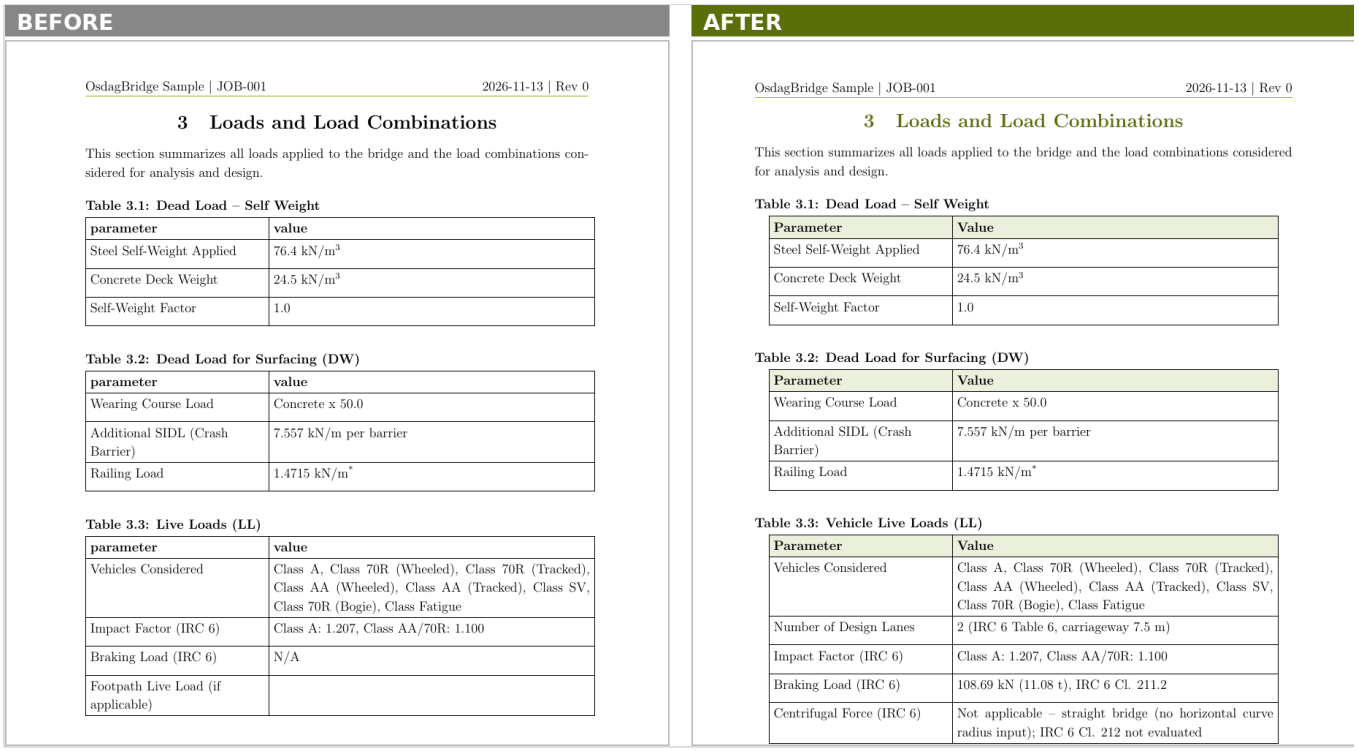


Figure 2 — Chapter 3 on page 12, before and after.

Results at a glance

Four new figures let a reviewer see the design outcome before reading any table. The original report had no graphs.

- **Readable without colour:** PASS and FAIL always carry a text label, and FAIL bars are hatched. The result reads correctly in greyscale print and for colour-blind readers.
- **Limit line:** a red dashed line marks UR = 1.0, with its label in the margin so it never overlaps a bar.
- **Matching type:** charts use the report's own font (Computer Modern) and state units on every axis and value.
- **Material quantities:** shown as values with their share of the total; the quantity bars use a darker brand green for contrast.
- **No dual axes:** concrete (m³) and reinforcement (MT) sit in two side-by-side panels rather than one chart with two y-axes, which would invite a false comparison.

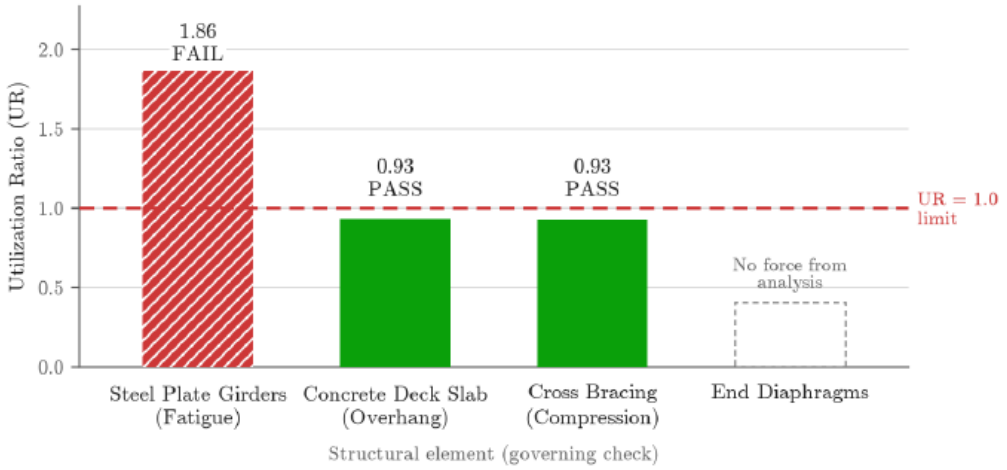


Figure 5.1: Overall utilization ratio of each structural element (governing check shown in brackets). Bars above the dashed line at UR = 1.0 fail.

Figure 3 — Utilization ratio by element (report Figure 5.1).

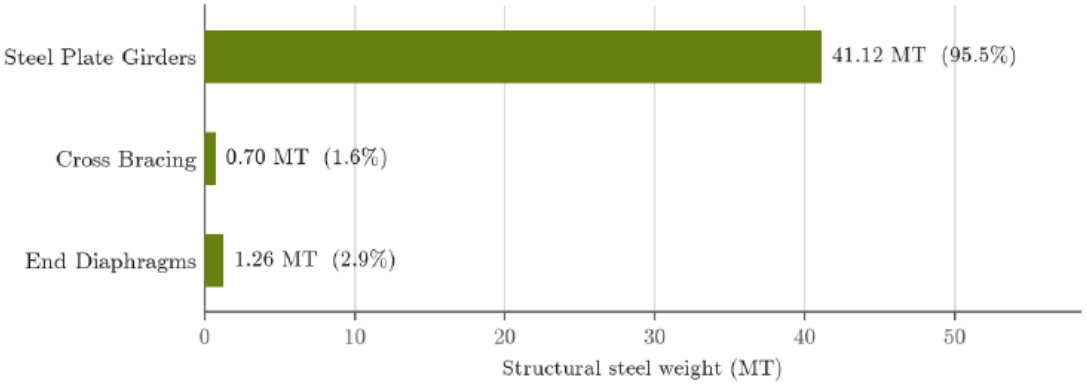


Figure 7.1: Structural steel weight by element (total of plotted elements: 43.08 MT). Values from Table 7.1.

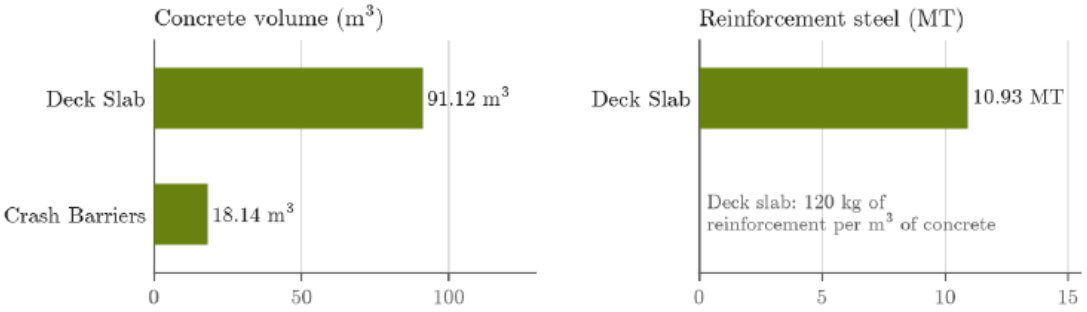


Figure 7.2: Concrete volume (m^3) and reinforcement steel weight (MT). Reinforcement is calculated for the deck slab only. Values from Table 7.1.

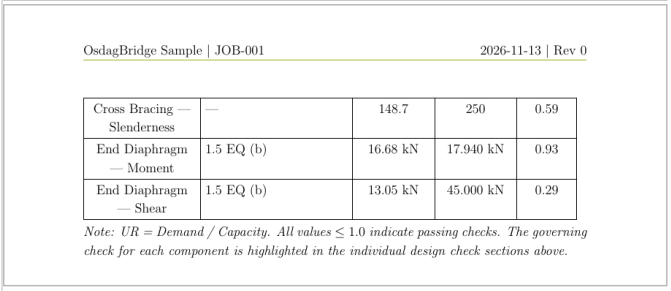
Figure 4 — Material quantities (report Figures 7.1 and 7.2).

No misleading values

Where the design has no result, the report now says so, instead of showing a zero or another member's numbers.

- **Before:** the End Diaphragm rows repeated the cross-bracing results exactly (16.68 kN demand, 17.940 kN capacity, UR 0.93), and Section 5.4's tables did the same.
- **After:** Section 5.4 explains that the analysis gave the end diaphragm no force. The summary row reads "No force from analysis", and the graph draws an empty dashed outline rather than a bar at 0.
- **Graphs and quantities:** a missing quantity shows as "Not calculated", and an element with any missing part is never partly summed.

BEFORE

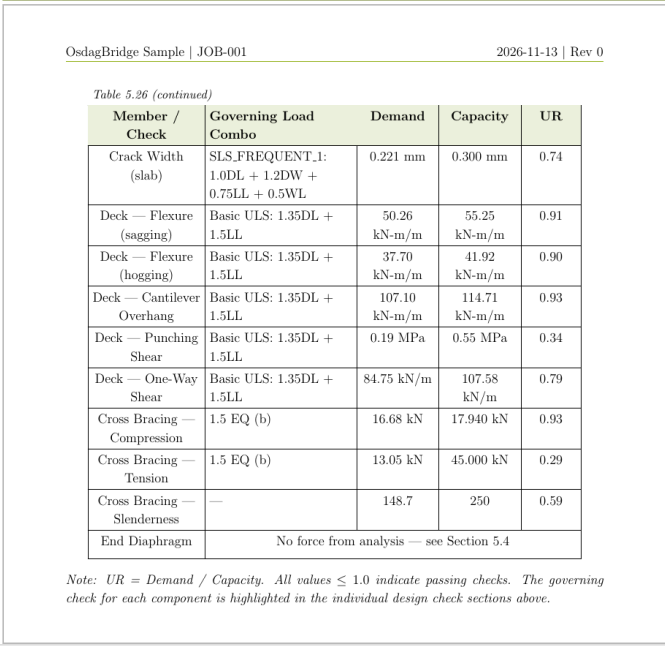


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Cross Bracing — Slenderness	—	148.7	250	0.59
End Diaphragm — Moment	1.5 EQ (b)	16.68 kN	17.940 kN	0.93
End Diaphragm — Shear	1.5 EQ (b)	13.05 kN	45.000 kN	0.29

Note: UR = Demand / Capacity. All values ≤ 1.0 indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.

AFTER



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Table 5.26 (continued)

Member / Check	Governing Load Combo	Demand	Capacity	UR
Crack Width (slab)	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	0.221 mm	0.300 mm	0.74
Deck — Flexure (sagging)	Basic ULS: 1.35DL + 1.5LL	50.26 kN-m/m	55.25 kN-m/m	0.91
Deck — Flexure (hogging)	Basic ULS: 1.35DL + 1.5LL	37.70 kN-m/m	41.92 kN-m/m	0.90
Deck — Cantilever Overhang	Basic ULS: 1.35DL + 1.5LL	107.10 kN-m/m	114.71 kN-m/m	0.93
Deck — Punching Shear	Basic ULS: 1.35DL + 1.5LL	0.19 MPa	0.55 MPa	0.34
Deck — One-Way Shear	Basic ULS: 1.35DL + 1.5LL	84.75 kN/m	107.58 kN/m	0.79
Cross Bracing — Compression	1.5 EQ (b)	16.68 kN	17.940 kN	0.93
Cross Bracing — Tension	1.5 EQ (b)	13.05 kN	45.000 kN	0.29
Cross Bracing — Slenderness	—	148.7	250	0.59
End Diaphragm	No force from analysis — see Section 5.4			

Note: UR = Demand / Capacity. All values ≤ 1.0 indicate passing checks. The governing check for each component is highlighted in the individual design check sections above.

Figure 5 — End-diaphragm rows of the summary table, before and after.

Page details

Small details on every page now behave correctly.

Detail	Before	After
Footer text colour	Green: the rule's colour leaked into the footer text	Black, with only the rule in green
Page numbering	Off by one (PDF page 21 read "Page 20 of 55")	Matches the PDF page on every page
Section headings at a page bottom	"5.5 Overall Design Check Summary" stranded at the foot of page 46	Kept with its intro text and table
Symbols such as ≤, ≥, ≠	Silently dropped by pdflatex in some places	Printed everywhere
"* Software default value" note	Pushed up into the text area, close to table rows	Sits in the footer zone, 28 pt or more below the text

BEFORE

Table 5.28: End Diaphragm Design — Capacity Summary

Panel	Member	Section	Governing LC	Demand (kN)	Capacity (kN)	UR	Status
G1-G2	Diagonal	40 x 40 x 3	1.5 EQ (b)	11.762	14.240	0.720	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	10.156	17.940	0.930	PASS
G2-G3	Diagonal	40 x 40 x 3	1.5 EQ (a)	7.150	45.000	0.160	PASS
	Chord	40 x 40 x 3	1.5 EQ (a)	6.174	45.000	0.140	PASS
G3-G4	Diagonal	40 x 40 x 3	1.5 EQ (a)	8.011	45.000	0.180	PASS
	Chord	40 x 40 x 3	1.5 EQ (a)	6.917	45.000	0.150	PASS
G4-G5	Diagonal	40 x 40 x 3	1.5 EQ (b)	13.109	45.000	0.290	PASS
	Chord	40 x 40 x 3	1.5 EQ (b)	11.319	45.000	0.250	PASS

Note: Designed per IS 800 Cl. 7 (compression) and Cl. 6 (tension). OsdagBridge cross-bracing module used.

5.5 Overall Design Check Summary

AFTER

5.4 End Diaphragm Design

End diaphragms at the supports transfer transverse loads to the bearings, restrain the bottom flanges against lateral displacement, and maintain the girder cross-section geometry during construction and in service. They are designed per IS 800:2007 and IRC 24:2010 Cl. 507.
No end-diaphragm design results: the structural analysis produced no force in the end-diaphragm members, because the analysis model restrains every girder end laterally at the supports and lateral load is taken directly by the supports. The end-diaphragm section is therefore not checked in this report; it is not a passing (UR = 0) result.

5.5 Overall Design Check Summary

Table 5.26 lists the governing load combination, demand, capacity and utilization ratio (UR = Demand / Capacity) of every design check. The same values are plotted in Figure 5.1 and Figure 5.2. The overall UR of a structural element is the highest UR among its checks (its governing check), so a failure of any one check is shown as a failure of the element.

Table 5.26: Overall Design Check Summary — All Members

Member / Check	Governing Load Combo	Demand	Capacity	UR
Girder — Moment	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	2652.52 kNm	11564.06 kNm	0.23
Girder — Shear	ACCIDENTAL_1: 1.0DL + 1.0DW + 0.75LL	467.52 kN	720.21 kN	0.65
Girder — LTB (constr.)	SW	220.64 kNm	900.22 kNm	0.25
Girder — Deflection	1.0 DL + 1.0 LL	26.78 mm	50.00 mm	0.54
Girder — Stress	SLS_RARE_1: 1.0DL + 1.2DW + 1.0LL + 0.6WL	96.01 MPa	315.00 MPa	0.30
Girder — Fatigue	SLS_FREQUENT_1: 1.0DL + 1.2DW + 0.75LL + 0.5WL	172.31 MPa	92.49 MPa	1.86
Transverse Shear (slab)	—	—	—	—

Continued on next page

Figure 6 — The stranded Section 5.5 heading (before, page 46) and the same section after (page 48).